Name:____ **Date:** ________________

Problem 1. *Find all the holes and vertical asymptotes of each of the following functions (hint: start by factoring):*

a) $f(x) = \dfrac{x}{x^2 - x}$

b) $f(x) = \dfrac{x^2 - 2x + 1}{x - 1}$

c) $f(x) = \dfrac{x + 7}{x^3 + 5x^2 - 14x}$

d) $f(x) = \dfrac{x^3 - 7x^2 + 12x}{x^4 - 5x^3 + 6x^2}$

Problem 2. *For each of the following piecewise functions, determine whether or not it has any jump discontinuities. For those which have jumps, tell me where they occur.*

***a*)** $f(x) = \begin{cases} x^2 + 3x - 5 & x \geq -2 \\ x^3 + 1 & x < -2 \end{cases}$

***b*)** $f(x) = \begin{cases} e^x + 1 & x > 1 \\ 1 - x^2 & x \leq 1 \end{cases}$

***c*)** $f(x) = \begin{cases} x^3 + 2x^2 - 3 & x \leq -3 \\ x - 9 & -3 < x \leq 0 \\ |3x + 2| & x > 0 \end{cases}$

Problem 3. *For each part, write the equation of a function that has holes and vertical asymptotes at the given points (you may leave it in a factored form!)*

***a*)** Holes at $x = 1, x = -2$; a vertical asymptote at $x = 0$

***b*)** A hole at $x = 3$; vertical asymptotes at $x = -5, x = \frac{2}{3}$